

# CS 486/686

# Neural Networks I

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Lecture 19

RN 21.1 · PM 7.5

# Learning goals

- Describe the **simple mathematical model** of a neuron.
- Describe desirable properties of an **activation function**.
- Distinguish **feedforward** from **recurrent** networks.
- Learn a **perceptron** for a simple logical function.
- Determine the logical function represented by a perceptron.
- Explain why a perceptron **cannot represent XOR**.
- Understand **recurrent neural networks**.

# Learning complex non-linear relationships

In vision, speech, translation, and beyond, the relationship between inputs and outputs is too complex to hand-program.

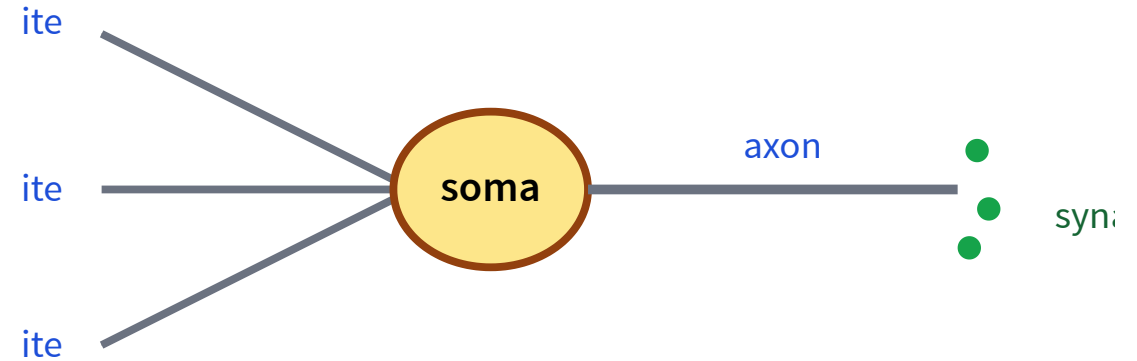
We need a model that can:

- Learn complex relationships.
- Be trained efficiently on lots of data.
- Generalize without overfitting.

Loose inspiration: the human brain — many simple components, densely connected.

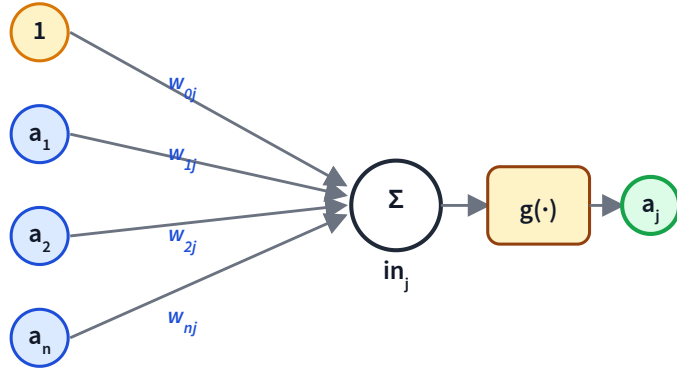
# From brain neurons to math neurons

- **Dendrites** receive input signals.
- **Soma** processes the signals.
- **Axon** sends an output signal.
- **Synapses** connect neurons together.
- Neurons fire (or don't) based on the inputs.



The math model abstracts this into **weighted sum + non-linear activation**.

# The math model (McCulloch & Pitts, 1943)



$$in_j = \sum_{i=0}^n w_{ij} a_i$$
$$a_j = g(in_j)$$

$a_0 = 1$  is the bias;  $w_{0j}$  acts as a threshold.  $g$  is a non-linear **activation function**.

# Desirable properties of the activation function

## Non-linear

Complex relationships are non-linear; stacking linear layers stays linear.

## Mimics real neurons

Fires ( $\approx 1$ ) when input is big; quiet ( $\approx 0$ ) otherwise.

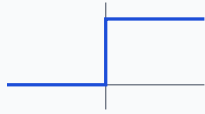
## Differentiable

So we can train via gradient-based optimization (next lecture!).

# Four common activation functions

## Step

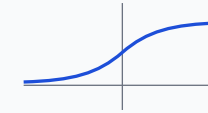
$$g(x) = 1 \text{ if } x > 0, \text{ else } 0.$$



Simple but *not differentiable*; rarely used in practice.

## Sigmoid

$$g(x) = \frac{1}{1 + e^{-x}}$$



Smooth, but *vanishing-gradient* in saturation regions.

## ReLU

$$g(x) = \max(0, x)$$



Fast, sparse; *dying-ReLU* when input stays negative.

## Leaky ReLU

$$g(x) = \max(0, x) + k \cdot \min(0, x)$$



Small negative slope keeps gradient alive on the left.

# Networks: feedforward vs recurrent

## Feedforward

Connections form a **DAG** (no loops). The network is just a function of its inputs.

## Recurrent

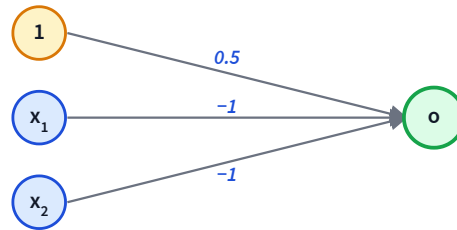
Outputs feed back as inputs. Supports **short-term memory** — behavior depends on history.

## Perceptron

A **single-layer feedforward** network: inputs connect directly to outputs. Can represent AND, OR, NOT and many other logical functions.

# Identify a perceptron

**Q1.** The perceptron below uses step activation with weights  $w_0 = 0.5$ ,  $w_1 = -1$ ,  $w_2 = -1$ . What logical function does it compute?



- A.  $x_1 \wedge x_2$
- B.  $\neg(x_1 \wedge x_2)$
- C.  $x_1 \vee x_2$
- D.  $\neg(x_1 \vee x_2)$  (NOR)

**D — NOR.** The output is  $g(0.5 - x_1 - x_2)$ , which is 1 only when  $0.5 - x_1 - x_2 > 0$ , i.e. both  $x_1$  and  $x_2$  are 0.

# Learn a perceptron: AND

**Q2.** Find weights  $(w_0, w_1, w_2)$  so the perceptron  $o = g(w_0 + w_1x_1 + w_2x_2)$  computes  $x_1 \wedge x_2$ . The AND truth table outputs 1 *only* on  $(1, 1)$ .

A.  $w_0 = -1, w_1 = 0.5, w_2 = 0.5$

B.  $w_0 = 0.5, w_1 = -1, w_2 = 1$

C.  $w_0 = 1.5, w_1 = -1, w_2 = -1$

D.  $w_0 = -1.5, w_1 = 1, w_2 = 1$

**D.** Check:  $-1.5 + x_1 + x_2 > 0$  iff  $x_1 + x_2 > 1.5$ , i.e. both equal 1. The other options either misclassify  $(1, 1)$  or fire on  $(0, 0)$ .

# More perceptron arithmetic

Step activation:  $g(x) = 1$  if  $x > 0$ , else 0.

**Q3.** What does  $h_1 = g(x_1 + x_2 - 0.5)$  compute?

- A.  $x_1 \vee x_2$  (OR)
- B.  $x_1 \wedge x_2$
- C.  $\neg(x_1 \vee x_2)$
- D.  $\neg(x_1 \wedge x_2)$

**A — OR.** Outputs 1 if at least one input is 1.

**Q4.** What does  $h_2 = g(-x_1 - x_2 + 1.5)$  compute?

- A.  $x_1 \vee x_2$
- B.  $x_1 \wedge x_2$
- C.  $\neg(x_1 \vee x_2)$
- D.  $\neg(x_1 \wedge x_2)$  (NAND)

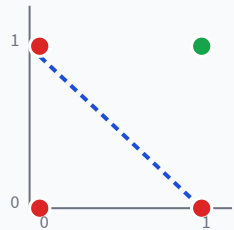
**D — NAND.** Outputs 0 only when both inputs are 1.

Remember  $h_1$  and  $h_2$  — we'll combine them to build XOR.

# Why a perceptron can't represent XOR

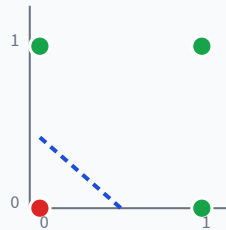
A perceptron is a **linear classifier**. Its decision boundary is a hyperplane in the input space.

**AND**



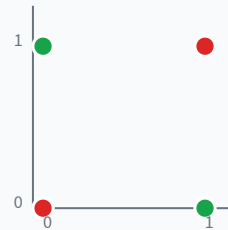
Separable

**OR**



Separable

**XOR**

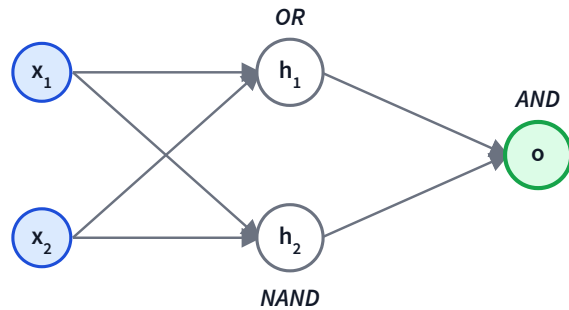


**Not separable**

Minsky & Papert (1969) showed this — and it triggered the first AI winter.

# XOR as a 2-layer network

Rewrite XOR using gates a perceptron *can* handle:  $x_1 \oplus x_2 = (x_1 \vee x_2) \wedge \neg(x_1 \wedge x_2) = h_1 \wedge h_2$ .



Truth-table check:

| $x_1$ | $x_2$ | $h_1$<br>(OR) | $h_2$<br>(NAND) | $h_1 \wedge h_2$ |
|-------|-------|---------------|-----------------|------------------|
| 0     | 0     | 0             | 1               | 0                |
| 0     | 1     | 1             | 1               | 1                |
| 1     | 0     | 1             | 1               | 1                |
| 1     | 1     | 1             | 0               | 0                |

Matches XOR exactly. One hidden layer is enough.

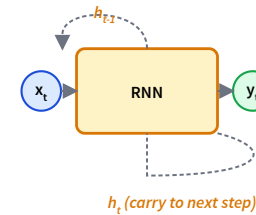
# Recurrent networks: process a sequence

When inputs come as a sequence (text, audio, sensor data), the same network reads one element at a time and carries a **state** across steps.

## Recurrence

$$h_t = f_W(h_{t-1}, x_t), \quad y_t = f_Y(h_t)$$

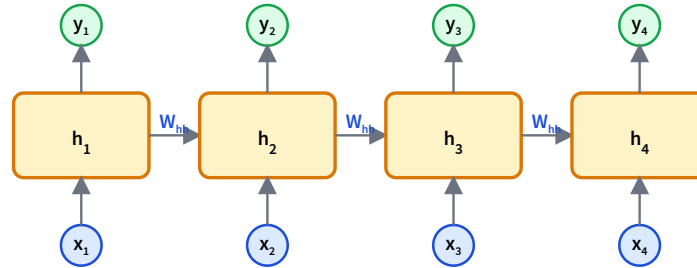
The state  $h_t$  summarizes everything seen so far.



Same weights at every step — the model can handle *any-length* sequence with a fixed parameter count.

# Vanilla RNN: equations + unrolled

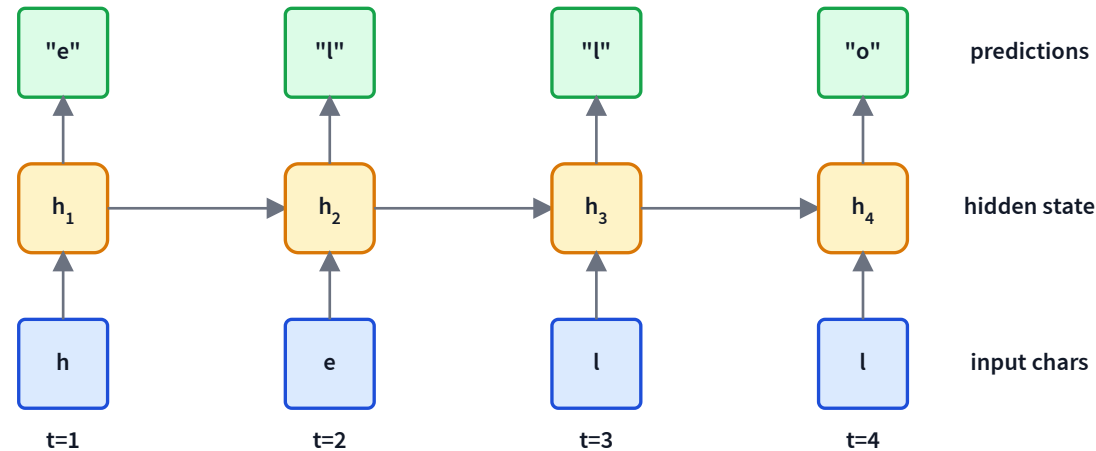
$$h_t = \tanh(W_{hh} h_{t-1} + W_{xh} x_t), \quad y_t = W_{hy} h_t$$



The **same**  $W_{hh}$ ,  $W_{xh}$ ,  $W_{hy}$  are reused at every time step.

# Application: language modeling

Feed characters one at a time; predict the next character at each step.



Trained to predict the next character; sample autoregressively to generate text.

# Learning goals (recap) — Next: training

- ✓ Describe the **math neuron**.
- ✓ Describe desirable properties of an **activation function**.
- ✓ Distinguish **feedforward** from **recurrent** networks.
- ✓ Learn / identify a **perceptron** for a logical function.
- ✓ Explain why a perceptron **cannot represent XOR**.
- ✓ Understand **RNNs** and how they process sequences.

L20: how do we actually *train* these networks? — gradient descent + **backpropagation**.